

**Molecular Assessment of Kallikrein 3 in Ameloblastoma and Odontogenic Keratocyst
(In-vitro Study)**

التقييم الجزيئي للكاليكرين في الورم الأرومي المينائي والتكيس الكيراتيني السني
(دراسة في المختبر)

Protocol

Submitted to Faculty of Dentistry, Cairo University

For Partial Fulfillment of the Requirement of the Doctoral Degree in Oral and Maxillofacial Pathology

By:

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I. Administrative information:

1. Title:

Molecular Assessment of Kallikrein 3 in Ameloblastoma and Odontogenic Keratocyst (In-vitro Study)

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5. Roles and responsibilities:

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II. Introduction:

6a. Statement of the Problem:

Ameloblastoma (AB), the most frequently encountered tumor arising from odontogenic epithelium, AB characterized by a benign but locally invasive behavior with a high risk of recurrence. The invasion of surrounding healthy tissues by tumor cells is one of the essential steps in tumor progression. Therefore, Identification of invasive activities in ameloblastomas may be useful to predict their biological behavior (Zhong, Guo et al. 2011). Odontogenic keratocyst (OKC) remains the most challenging to treat because of its locally-aggressive behaviour and high recurrence potential. OKC was previously reclassified a neoplasm by the 2005 World Health Organization (WHO) Working Group and renamed keratocystic odontogenic tumour. (Slootweg, Chan et al. 2017).

Tumor cell invasion is found to be enhanced in-vitro by KLK 1, 3, 10 and 13. KLK 6 also increases E-cadherin shedding and promotes cell proliferation, migration, and invasion. KLK could also cause proliferation of tumor cells. Generally, KLK overexpression in malignant tumors has, paradoxically, been shown to be associated with both favorable and poor prognoses. KLK 4 is known to have a role in the formation of normal tooth enamel, where it functions to degrade enamel proteins during maturation (Simmer, Richardson et al. 2011).

PICO:

Population1: Archival blocks of tissue specimens of ameloblastoma.

Population2: Archival blocks of tissue specimens of odontogenic keratocyst.

Indicator 1: Immunoexpression of kallikrein 3 on ameloblastoma and odontogenic keratocyst.

Indicator 2: Detection of kallikrein 3 gene mRNA on ameloblastoma and odontogenic keratocyst.

Control: Archival blocks of tissue specimens of dentigerous cyst.

Outcome: Protein and mRNA gene expression of kallikrein 3 in ameloblastoma and odontogenic keratocyst

Outcome name	<i>Measurement</i>	<i>Measuring Device</i>	<i>Measuring Unit</i>
Protein expression	Kallikrein 3 Immunoexpression (Area percent)	Computer image analyze	μm^2
Gene expression	Kallikrein 3 using Real time -PCR	Rotorgene PCR system	Fold change FLD=e

Research question:

Does kallikrein 3 have a definite role in the pathogenesis of ameloblastoma and odontogenic keratocyst?

6b. Review of literature:

In this study, Pubmed, Cochrane database of systematic reviews and Google Scholar were used as data bases.

Odontogenic tumours are lesions resulting from epithelium, ectomesenchymal and/or mesenchymal elements, which participate in the formation of the tooth structure. According to the World Health Organisation (WHO), these tumours are exclusively found in the maxillofacial region and they can occur in any age group (**Barnes, Eveson et al. 2005**)

Ameloblastoma is the most frequent tumor arising from odontogenic epithelium. It is characterized by a benign, but locally invasive behavior with a high risk for recurrence. It tends to infiltrate between the cancellous bones(**Sathi, Tamamura et al. 2012**), because of this features, partial maxillectomy or mandibulectomy with a safety margin of healthy bone is the most common choice of the treatment options(**Sachs 2006**) . According to the 2005 histological classification of tumors of the World Health Organization, ABs are classified into the following four variants: solid/multicystic which is the most common type, extraosseous/peripheral, desmoplastic and unicystic (**Barnes et al, 2005**). The most aggressive clinical and pathologic association is seen in the solid/multicystic type, which is associated with the highest recurrence rate of up to 90 % with conservative operations such as enucleation and curettage (**Rosenstein, Pogrel et al. 2001, Carlson and Marx 2006**)

The odontogenic keratocyst (OKC) remains the most challenging to treat because of its locally-aggressive behaviour and high recurrence potential. OKC was previously reclassified a neoplasm by the 2005 World Health Organization (WHO) Working Group and renamed keratocystic odontogenic tumor. However the 2017 WHO Consensus Panel's decision to re-instate it as an odontogenic cyst attests to our lack of understanding of its biological nature (**Slootweg, Chan et al. 2017**) .

OKC has an aggressive growth potential and a tendency to recur. It has a thin fibrous capsule with a lining of parakeratinized friable squamous epithelium which has a high mitotic activity and the remnants of it tend to be left on removal of the lesion contributing to the recurrence of the lesion (**Mendes, Carvalho et al. 2010**). Treatment of OKC is most often by enucleation or by surgical resection for large lesions(**El-Naggar, Chan et al. 2017**). A systematic review found an overall recurrence rate of about 25% after enucleation, while less than 2% after resection (**Johnson, Gannon et al. 2014**)

Human tissue kallikreins (KLK) are a group of 15 single-chain serine proteases with their genes (KLK 1-KLK 15) encoded on chromosome 13 of the human genome **(Emami and Diamandis 2007)**. Different subtypes of KLK are expressed in a wide variety of human tissue types in normal and pathological states. KLK is also secreted in the biologic fluid, such as seminal plasma, serum, and milk. In addition, multiple KLK can be also expressed in single tissue types, such as the salivary glands. Of all the KLK, only KLK 4 is known to have a role in the formation of normal tooth enamel, where it functions to degrade enamel proteins during maturation **(Simmer, Richardson et al. 2011)**.

However, human tissue KLKs are aberrantly expressed in multiple malignancies including those of the prostate, breast, and Ovary **(Emami and Diamandis 2007)**. Tumor cell invasion is found to be enhanced in-vitro by KLK 1, 3, 10 and 13. KLK 6 also increases E-cadherin shedding and promotes cell proliferation, migration, and invasion. KLK could also cause proliferation of tumor cells. Generally, KLK overexpression in malignant tumors has, paradoxically, been shown to be associated with both favorable and poor prognoses. **(Simmer, Richardson et al. 2011)**.

KLK 3, also known as prostate-specific antigen, is the most researched and best understood of the KLK family. It was investigated in different diseased conditions as well as in normal physiology **(Hollenberg 2014)**. KLK 3 has been shown to be involved in tumor invasion by direct extracellular matrix degradation, tumor activation by transforming growth factor- β and insulin-like growth factor, and in the growth of these lesions through well-known mechanisms and signalling cascades or through alternative mechanisms that have not yet been identified **(Darling, Woodford et al. 2017)**.

6c. The objective:

The current study aims to investigate the molecular assessment of Kallikrein-3 in ameloblastoma and odontogenic keratocyst. The results of this study will shed more light on the biologic behaviour of these lesions which will aid in the diagnosis and treatment of such lesions.

III. Methods

A) Samples, intervention and outcomes

7. Calculated sample size

According to (Jamshidi, Zargaran et al. 2014)

The largest value has Mean 14.47 SD 3.8 and the smallest value has Mean 9.6 SD 2.9. The maximum difference between the three groups is $(14.47 - 9.6) = 4.87$ and the standard deviation = 3.3 the size Effect = 1.475. Assuming the alpha error is 0.05 and the power 80%.

To consider possible drop out 11 samples per group is considered a total of 33.

Minitab computer output: One-way ANOVA

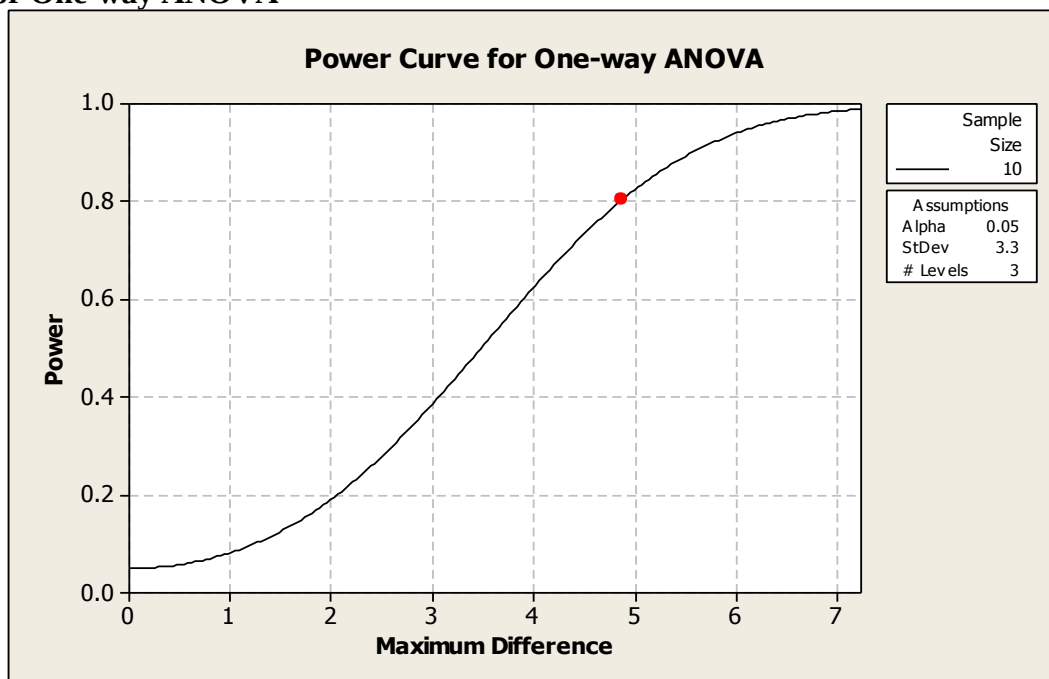
Alpha = 0.05 Assumed standard deviation = 3.3

Factors: 1 Number of levels: 3

Maximum Difference	Sample Size	Target Power	Actual Power
4.87	10	0.8	0.803884

The sample size is for each level

Power Curve for One-way ANOVA



8. Description of study sample

The archives of the Oral & Maxillofacial Pathology Department, Faculty of Oral and Dental Medicine, Cairo University will be reviewed for the last 5 years from the period of 2013-2018 to collect AB, OKC, dentigerous cyst

9. Intervention for each group

1. The collected blocks will be categorized into 3 groups: dentigerous cyst (comparator), OKC and AB.
2. The blocks will be serially cut into 5 µm thick sections and stained with hematoxylin and eosin to confirm the diagnosis.
3. Immunohistochemical staining of KLK 3.
4. PR- PCR technique of mRNA KLK3 gene

10. Outcomes

Area percent of immunohistochemical expression and Gene expression of the KLK3 in the 3 lesions.

Outcome name	<i>Measurement</i>	<i>Measuring Device</i>	<i>Measuring Unit</i>
Protein expression	Kallikrein-3 Immunoexpression (Area percent)	Computer image analyze	µm ²
Gene expression	Kallikrein-3 using Real time -PCR	Rotorgene PCR system	Fold change FLD=e

Steps for immunostaining:

- Sections will be deparaffinized by incubating them in xylene three times, 5 minutes each.
- Slides will be dehydrated in two changes of 100 % ethanol, 95% and 80% ethanol for 3 minutes each.
- Endogenous peroxidase activity will be blocked by immersing the slides in 3 % hydrogen peroxide solution 10 minutes at room temperature. Then will be washed two times with phosphate buffered saline (PBS), 3 minutes each.
- Slides will be dipped in antigen retrieval solution (10mM citrate buffer, PH 6.0) and will be heated to near boiling (95-100°C) for 10-20 minutes. Then, slides will be cooled in the buffer at room temperature for 20 minutes.
- The slides will be rinsed in PBS three times, 1 minute each.
- One or two drops of the ready to use prediluted primary antibody KLK 3 will be added to each section, slides were incubated for 30 minutes in a humid chamber to reduce reagent evaporation. Then, they will be washed in PBS three times, 3 minutes each.

- One or two drops of the biotinylated secondary antibody will be applied to each section and incubated for 10-15 minutes at room temperature. Then, rinsed with PBS three times, 3 minutes each.
- Two drops of streptavidin peroxidase will be added to each section and incubated for 10-15 minutes at room temperature. Then, rinsed with PBS three times, 3 minutes each.
- One or two drops of the DAB substrate chromogen (1-2 drops of DAB chromogen will be added to 1ml of DAB substrate) will be applied to each section and slides were incubated at room temperature for 10 minutes. Followed by washing in PBS for three minutes.
- The slides will be immersed in haematoxylin bath for 1-4 minutes, and then they will be washed.

Immunohistochemical interpretation:

- Immunohistochemical expression of KLK3 will be examined and measured as an area percent by the image analyzer computer system using the software Leica Quin 500.
- The most homogenous immunostained areas will be chosen for evaluation in standard measuring frame using a magnification of x400.
- Area percent from each slide will be obtained from 3 different fields.

RT-PCR procedure:

- Five Sections of 5 microns thick (25 μ) will be cut from each paraffin block and placed in an eppendorf tube.
- The commercial QIAamp RNA Mini Kit (Qiagen, Germany) was used for RNA extraction. The tissues will be dewaxed with xylene, followed by two washes with 100% ethanol twice to remove residual xylene.
- 240 μ l of Buffer PKD will be added, and mixed by vortexing.
- Centrifugation will be done for 1 min at 10,000 $\times g$.
- 10 μ l KLK3 will be added to the lower part, clear phase. Mix for the cocktail was done gently by pipetting up and down.
- Incubation will be done at 56°C for 15 min, then at 80°C for 15 min.
- The 15 min incubation at 80°C will be critical for reversal of cross-links and optimal RNA performance in downstream applications such as real-time RT-PCR.
- The lower solution will be transferred into a new 2 ml microcentrifuge tube.
- Incubation on ice for 3 min. will be done then, centrifugation for 15 min at 10,000 $\times g$ will be done.
- The supernatant will be transferred to a new microcentrifuge tube without disturbance of the pellet (The pellet contains insoluble tissue debris, including cross linked DNA).
- DNase Booster Buffer will be added which is equivalent to a tenth of the total sample volume (approximately 16 μ l or 25 μ l) and 10 μ l DNase I stock solution. The tube will be mixed by inversion.
- Incubation will be done at room temperature for 15 min.
- 320 μ l of Buffer RBC will be added to adjust binding conditions, and the lysate will be mixed thoroughly.
- 720 μ l ethanol (100%) will be added to the sample and mixed well by pipetting.
- Precipitates may be visible after addition of ethanol.
- 700 μ l of the sample will be transferred, including any precipitate that may form, to an RNeasy MinElute

spin column placed in a 2 ml collection tube (supplied). The lid will be closed gently, and centrifuged for 15 s at $\geq 8000 \times g$. The flow-through will be discarded.

- The previous step will be repeated until the entire sample passed through the RNeasy MinElute spin column.
- 1500 μl Buffer RPE will be added to the RNeasy MinElute spin column. The lid was closed gently, and centrifuged for 15 s at $\geq 8000 \times g$. The flow-through was discarded.
- 500 μl Buffer RPE will be added to the RNeasy MinElute spin column. The lid will be closed gently and centrifuged for 2 min at $\geq 8000 \times g$ to wash the spin column membrane. The collection tube with the flow-through will be discarded.
- The RNeasy MinElute spin column will be placed in a new 2 ml collection tube and centrifuged at full speed for 5 min. The collection tube with the flow-through was discarded.
- The RNeasy MinElute spin column will be placed in a new 1.5 ml collection tube. 14–30 μl RNase-free water will be added directly to the spin column membrane. The lid will be closed gently and centrifuged for 1 min at full speed to elute the RNA.
- Real-time quantitative PCR will be performed using SYBR Green Universal Master Mix (2X) (Applied Biosystems, Warrington, WA1 4SR, UK) according to the manufacturer's protocol. Cycle conditions will be as following: 95° C for 10 min followed by 45 cycles (95° C denaturation for 15 s, 60° C annealing for 1 min, 72° C extension for 1 min), with a final incubation at 72° C for 5 min.
- A relative quantification method ($2^{-\Delta\Delta C_T}$ method will be used, where C_t is cycle threshold) will be chosen for quantitative analysis. The relative quantitation value of target will be normalized to the endogenous control GAPDH (housekeeping) gene.

B) Assignment to intervention

This study is non-randomized.

11. Sequence generation

Can't be applied.

12. Allocation concealment

Can't be applied.

13. Implementation

Can't be applied.

C) Blinding

14. Blinding

Can't be applied.

D) Statistical methods

The collected data will be presented as mean and standard deviation and statistically analysed. ANOVA test followed by *Tukey's post-hoc test* for pair-wise comparison between the studied groups will be used.

IV- Ethics:

16. Research ethics approval

V-Statement of originality

17. Statement of originality:

- Only one study assessed the expression of KLK3 in OKC and AB. However, the results in this study were not clear enough as the mean and SD were not reported. Moreover, the author didn't mention the number of cases used in the study. It was not also clarified if KLK3 has a role in the aggressive behaviour of AB and OKC or not (Darling, Woodford et al. 2017). Up to our knowledge the expression of KLK3 on aggressive behaviour of AB and OKC need further investigation.
- The current study will shed more light about the role of KLK3 in invasion potential, and aggressiveness of these lesions and that may help in early diagnosis and better treatment for these lesions.

References:

- Barnes, L., J. Eveson, P. Reichart and D. Sidransky (2005). World Health Organization classification of tumours. Pathology and genetics of head and neck tumours, Lyon: IARC Press.
- Bustin, S. (2002). "Quantification of mRNA using real-time reverse transcription PCR (RT-PCR): trends and problems." Journal of molecular endocrinology **29**(1): 23-39.
- Carlson, E. R. and R. E. Marx (2006). "The ameloblastoma: primary, curative surgical management." Journal of oral and maxillofacial surgery **64**(3): 484-494.
- Darling, M. R., R. Woodford, K. K. Cuddy, L. Jackson-Boeters, A. Hayter, J. Inkaran, E. P. Diamandis and Z. Khan (2017). "Kallikrein-related peptidase expression in odontogenic cysts and tumors: An immunohistochemical comparative study." Journal of investigative and clinical dentistry **8**(4): e12256.
- El-Naggar, A., J. Chan, J. Grandis, T. Takata and P. Slootweg (2017). "Tumours of the oral cavity and mobile tongue." World Health Organization (WHO) classification of head and neck tumours. International Agency for Research on Cancer (IARC). 4th ed. Lyon: IARCPress: 105-115.
- Emami, N. and E. P. Diamandis (2007). "Human tissue kallikreins: a road under construction." Clinica chimica acta **381**(1): 78-84.
- Guénin, S., M. Mauriat, J. Pelloux, O. Van Wuytswinkel, C. Bellini and L. Gutierrez (2009). "Normalization of qRT-PCR data: the necessity of adopting a systematic, experimental conditions-specific, validation of references." Journal of experimental botany **60**(2): 487-493.
- Hollenberg, M. D. (2014). "KLKs and their hormone-like signaling actions: a new life for the PSA-KLK family." Biological chemistry **395**(9): 915-929.
- Jamshidi, S., M. Zargarani, F. Baghaei, S. Shojaei, R. Z. Mahmoodabadi, A. Dehghan and A. Moghimbeigi (2014). "An immunohistochemical survey to evaluate the expression of CD105 and CD34 in ameloblastoma and odontogenic keratocyst." Journal of Dentistry **15**(4): 192.
- Johnson, N. R., O. M. Gannon, N. W. Savage and M. D. Batstone (2014). "Frequency of odontogenic cysts and tumors: a systematic review." Journal of investigative and clinical dentistry **5**(1): 9-14.
- Macdonald-Jankowski, D. (2010). "Orthokeratinized odontogenic cyst: a systematic review." Dentomaxillofacial Radiology **39**(8): 455-467.
- McDonald, J. S. (2016). Tumors of the oral soft tissues and cysts and tumors of bone. McDonald and Avery's Dentistry for the Child and Adolescent (Tenth Edition), Elsevier: 603-626.
- Mendes, R. A., J. F. Carvalho and I. van der Waal (2010). "Characterization and management of the keratocystic odontogenic tumor in relation to its histopathological and biological features." Oral oncology **46**(4): 219-225.
- Rosenstein, T., M. A. Pogrel, R. Smith and J. A. Regezi (2001). "Cystic ameloblastoma—behavior and treatment of 21 cases." Journal of oral and maxillofacial surgery **59**(11): 1311-1316.
- Sachs, S. A. (2006). "Surgical excision with peripheral ostectomy: A definitive, yet conservative, approach to the surgical management of ameloblastoma." Journal of oral and maxillofacial surgery **64**(3): 476-483.
- Sathi, G. A., R. Tamamura, H. Tsujigiwa, N. Katase, M. Lefeuve, C. H. Siar, H. Matsuda and H. Nagatsuka (2012). "Analysis of immunoexpression of common cancer stem cell markers in ameloblastoma." Experimental and therapeutic medicine **3**(3): 397-402.
- Simmer, J. P., A. S. Richardson, C. E. Smith, Y. Hu and J. C. C. Hu (2011). "Expression of kallikrein-related peptidase 4 in dental and non-dental tissues." European journal of oral sciences **119**: 226-233.
- Slootweg, P., J. Chan, E. Stelow and L. Thompson (2017). "Tumours of the nasal cavity, paranasal sinuses

and skull base." WHO classification of head and neck tumors. 4th ed. Lyon: IARC: 14-27.

- Zhong, Y., W. Guo, L. Wang and X. Chen (2011). "Molecular markers of tumor invasiveness in ameloblastoma: An update." Annals of maxillofacial surgery **1**(2): 145.